Tutorial 3: Climate Data Analysis

Pushpendra

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Introduction

In this tutorial, we explored how global near-surface air temperature has evolved over time using reanalysis datasets. The primary objective was to preprocess raw climate data, construct spatial temperature maps, and investigate long-term trends in warming and cooling across the globe. Additionally, we examined the statistical significance of these observed changes to distinguish robust climate signals from regions dominated by variability.

Data and Preprocessing

The dataset used was `air.2m.gauss.1948_2024_noleap_merged.nc`, containing monthly mean 2 m air temperature values on a Gaussian grid spanning 1948–2024.

The preprocessing steps involved:

- Converting temperatures from Kelvin (K) to Celsius (°C).
- Aggregating monthly values into annual mean temperatures to suppress short-term variability.
- Defining a climatological baseline (1951–1980) and computing annual anomalies as deviations from this baseline mean.

Spatial Temperature Maps

Two kinds of spatial maps were generated for each year:

1. **Absolute Temperature Maps**: Show annual mean near-surface air temperature in °C. These provide a direct sense of the Earth's climate state in a given year.
2. **Anomaly Maps**: Display deviations from the baseline (1951–1980), using a red–blue color scale. Warm anomalies (red) indicate higher-than-baseline years, while blue indicates cooler years.

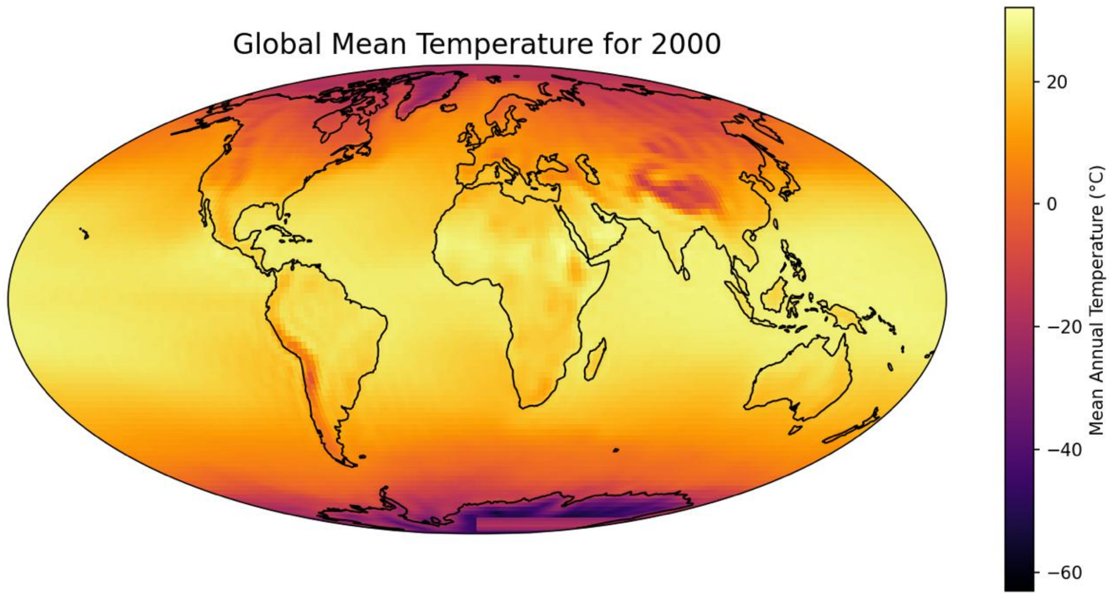

Figure 1: Sample global absolute temperature map.

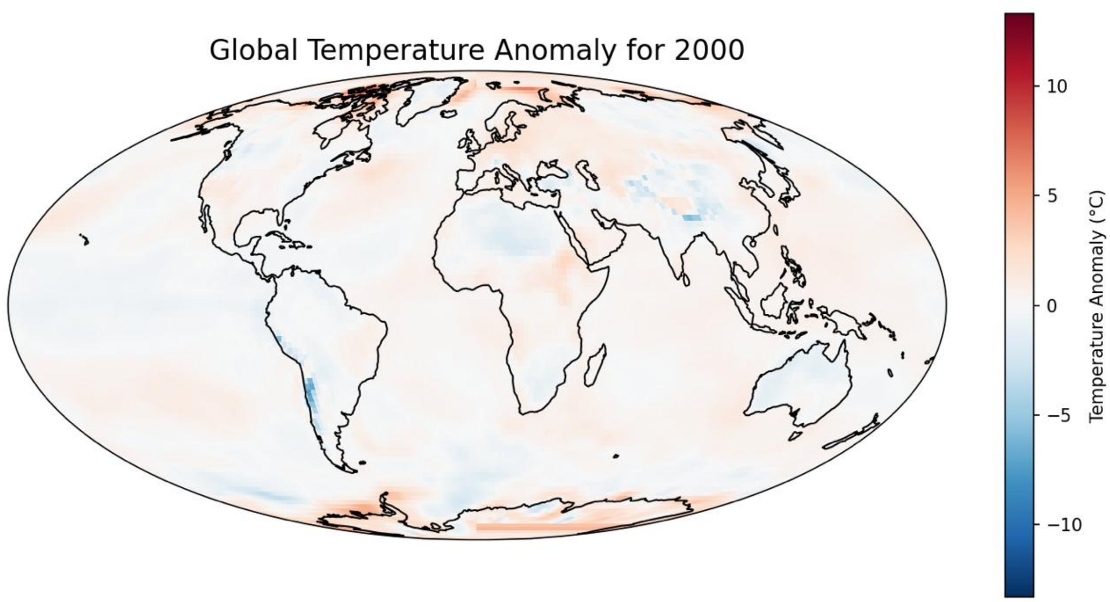

Figure 2: Global anomaly map relative to the 1951–1980 baseline.

Long-Term Climate Trends

To quantify how climate has changed, we conducted a grid-cell-wise linear regression on the annual mean temperature time series. This yielded:

- **Trend Magnitude (°C/decade)**: The slope of warming or cooling across the period.
- **Trend Significance (p-value)**: Indicates whether the observed trend is statistically

significant (using a 5% threshold).

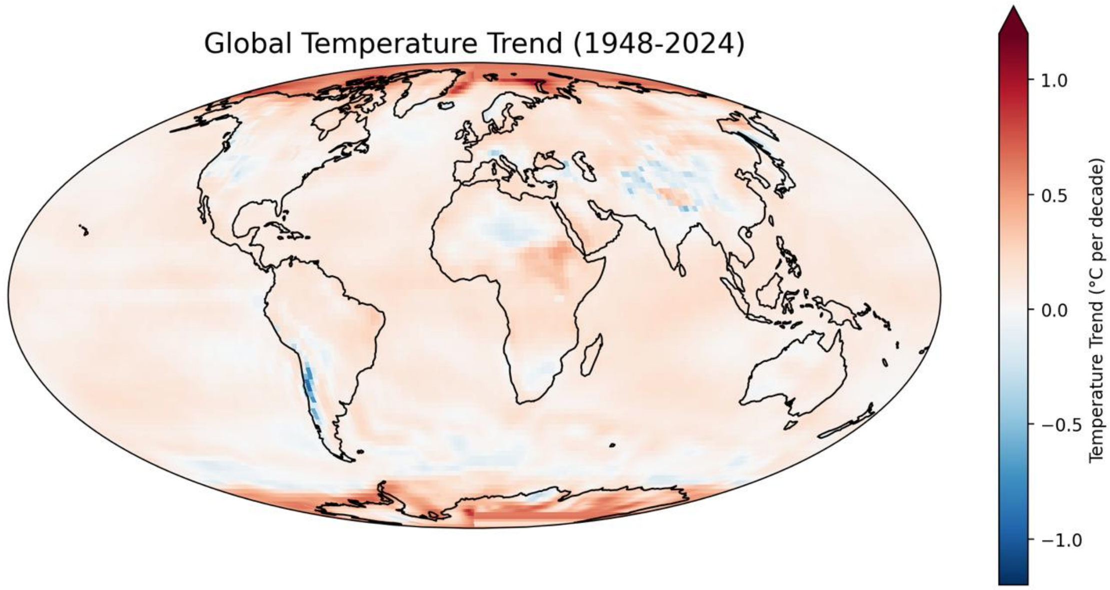

Figure 3: Spatial distribution of temperature trends (°C/decade).

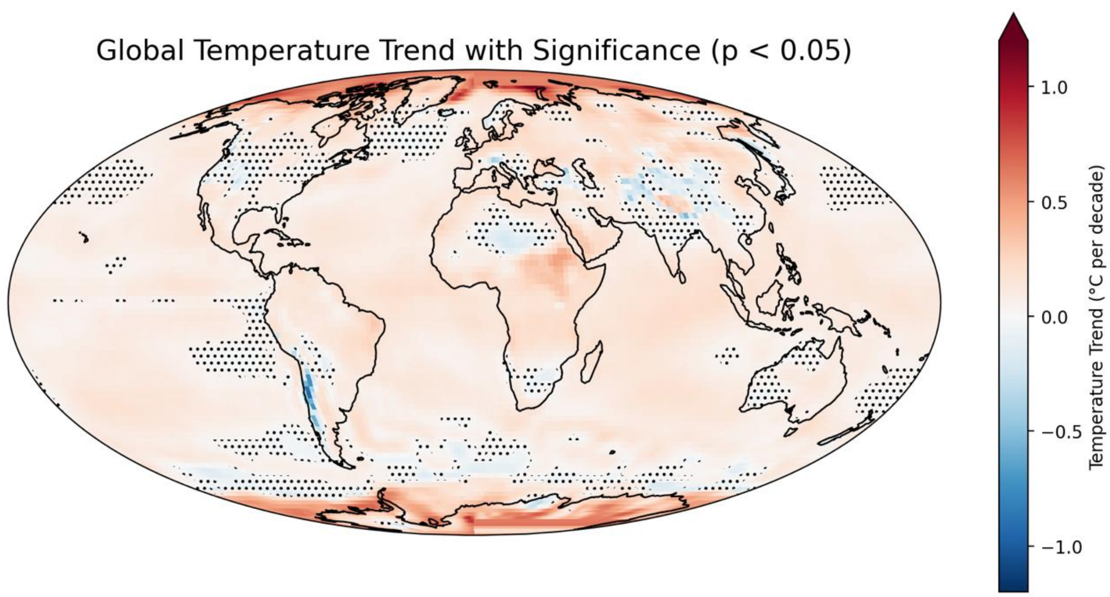

Figure 4: Trend significance map with stippling in regions where trends are not statistically significant ($p \geq 0.05$).

Observations and Insights

Regions of Strong Warming

- **Arctic and high northern latitudes**: The largest positive trends are observed here. This is consistent with *Arctic amplification*, where polar regions warm at a rate faster than the global average.
- **Antarctic Peninsula and parts of the Southern Ocean**: These regions also exhibit marked warming signals.

Regions of Cooling

- **Western South America**: A narrow but intense cooling signal is present along the Andes.
- **Localized regions in China and India**: Small but noticeable cooling pockets appear, though spatially limited.

Regions with Non-Significant Trends

- Large stretches of the open ocean, particularly the subtropical gyres, much of the central Pacific, and parts of the Atlantic and Indian Oceans, show dense stippling—indicating weak or statistically insignificant trends.
- Over India, monsoon-driven variability leads to non-significant results in most grid cells.
- Central and eastern North America also show many non-significant local trends, likely reflecting strong decadal variability linked to modes such as the PDO and NAO.

Conclusion

This exercise highlights the power of combining reanalysis datasets with simple statistical tools to extract meaningful climate signals. Key takeaways include:

- Global warming is spatially heterogeneous, with polar regions showing the strongest amplification.
- Cooling patches exist but are limited and often localized.
- Large oceanic and monsoon-affected regions exhibit high variability, making long-term signals statistically weaker.